Learning Money by Living It

A pixel-art life simulation that teaches financial literacy by letting students live it

Team No.: 97

M2 — Progress report

Status at 03:30, 11 Oct. All three life stages are playable end to end in a phone-sized browser. The game engine is unit-tested and driven by scripted players; the content pack (24 tasks $\times$ 3 personas, Telugu) validates against its schema; the app installs as an offline PWA; the report service emails a PDF to a sandbox inbox. Code is on `main` (68 incremental commits from four contributors).

1. Current Approach and Changes Since M1

Unchanged: three-stage life simulation, Pokémon-style money battles, teach-after-the-choice with abstract animations, mistake bank, persona framing, difficulty buckets, before/after quiz, emailed report, NGO join codes, the guardrail.

Decided since M1 (M1 left these open):

- **Economy.** Everyone starts Stage 1 with $S = ₹5{,}000$ in-game per month; in-game price $=$ real price $\times S/$ monthly pocket money, with pocket money taken from the indirect weekly-spend band (₹200, ₹800, ₹2,000, ₹4,000). Stages 2–3 use fixed reference incomes (₹20,000 and ₹50,000 real $\rightarrow$ ₹25,000 and ₹60,000 in-game).
- **Time.** Ten monthly chapters: 4 (Intermediate) $+$ 3 (first salary) $+$ 3 (earning more), then the epilogue at 60. Stage jumps skip 5, 6 and 25 years, so compounding on the four made-up assets is visible.
- **Happiness.** 0–100, starts at 70, decays 3 per chapter; choices move it $-30 \ldots +15$; zero restarts the task; below 30 locks mini-games.
- **Mini-games.** Two, skippable: *Kirana Rush* (sort needs from wants) and *Scam or Legit* (swipe fake messages). A win pays 8% of monthly income as a side gig and $+10$ happiness.
- **Guide and hints.** A guide character delivers the lesson line; hints switch on after two failures of the same task.
- **Languages.** Telugu for the whole UI (158/159 keys) and for onboarding, quiz, persona cards and two demo tasks; a Hindi file with only the title and buttons proves "add a language $=$ add a file" with English fallback.
- **Stack.** TypeScript + Phaser 3 + Vite (PWA), Vitest, zod schema validation; a small Node report service; a static NGO dashboard mock. One generic battle screen is driven entirely by JSON content.

2. Experiments Run So Far and Results

Economy scaler check (unit-tested): a ₹200 cricket bat costs ₹200 / ₹400 / ₹1,000 in-game for students with ₹5,000 / ₹2,500 / ₹1,000 of monthly pocket money, i.e. the same share of their money as in real life.

Scripted players on the real content (20 seeds per cell, same seeds across cells). Three policies play all ten chapters: *always good*, *random*, *always worst* (which never repeats a fatal choice on a restart, so it can finish).

Policy	Difficulty	Happiness end / min	Portfolio at 60	Detours	Restarts	Went broke	Literate	Finished
good	all three	98 / 66	₹1.3–2.0 L	0	0	0%	100%	100%
random	all three	34–64 / 25–51	₹0.9–1.3 L	1.0–1.6	0.7–1.2	10–30%	0–10%	100%
worst	all three	7–35 / 4–32	0	2.4–3.9	2.2–4.0	20–90%	0%	100%

Reading: good play ends stable and literate; careless play detours 1–2 times and sometimes goes broke; the worst play restarts several times, can go broke (an uninsured ₹2 lakh hospital bill), yet always finishes. The first run of this table showed *good* play scoring literate only 20–80% and nobody ever going broke; both were fixed (NCFE evidence rule; a genuinely expensive trap) and re-measured.

Unit tests: 124 passing — engine 68 (RNG reproducibility, scaler, delayed/repeating effects, task selection with pillar rotation and the investing gate, restart/detour routing, mistake bank, investing returns, NCFE scoring, what-if determinism, save/resume round-trip), content 24, mini-games 20, game helpers 11, report 1.

Content validation: 24 main tasks (12 / 6 / 6 per stage), 72 persona framings, 9 detours, 16 random events, 17 explainer animations, 17 concepts, 38 delayed or repeating effects; every NCFE item (17/17) has at least one tagged choice; every cross-reference (concept, explainer, detour, place, event, flag) resolves; a double-blind string check passes.

Devices and network (production build, headless Chromium, 360$\times$640): Fast 3G + 4$\times$ CPU throttle: **5.1 s** to first frame on first visit, **1.4 s** on a repeat visit; Slow 3G: 20.5 s (538 KB transferred). **Offline reload: 0.5 s** via the service worker. The M1 target of 5 s held only on Fast 3G; on Slow 3G the bundle must shrink (asset trimming cut the precache from 1,638 to 65 files; Phaser itself is the floor).

Report pipeline: a seeded game produced a PDF report emailed to a sandbox inbox with the guardian in CC; with no network the service stores the message locally and the app offers a download instead.

Play test (team, 02:15): four players took the build from title to Stage 3 on a laptop in phone mode and on a phone, unguided. **24 issues logged** (6 blocking, 14 affecting the teaching or demo path, 4 polish), each with an owner. The findings that change the design: (i) the economy is too forgiving — careless play still ends with a large balance, so weak and critical choices will cost more and stage incomes drop; (ii) several choices and mini-game items are

ambiguous (a deodorant can be a need), so every task will be re-read against a pinned concept list and the needs-vs-wants game restricted to unarguable items; (iii) nobody understood the hearts without a label, and the quiz screen needs a heading; (iv) the 9:16 canvas letterboxes on modern phones; (v) the scam-spotting mini-game's 30-second limit is too short. Also wanted: step sounds and a real walk, better avatars offered by sex, a recognisable guide character, Hindi alongside Telugu, read-aloud options. All 24 are tracked in the repository (`docs/plan/PLAYTEST-01.md`); a second play test follows the fixes at 07:00.

3. Progress on Testing

Learning gain	Before/after quiz wired end to end; NCFE-aligned score (8 knowledge, 9 behaviour, 3 attitude) and safety score computed from tagged choices	in place
Usability	Team play test done (24 issues, owners assigned); second play test after fixes	02:15 done; 07:00 next
Engine correctness	124 unit tests; scripted-player table above; save/resume round-trip; seeded what-if replay	done, repeating
Devices & network	Throttled timings and offline reload measured on the production build	done
Language	Validator checks every Telugu key exists in English (158/159); Telugu onboarding screenshots	keys done; screenshots pending
Report & privacy	Sandbox-inbox send verified; nickname-only state; the app never asks for OTP/PIN/Aadhaar	done

4. Plan for the Remaining Time

- **04:00–08:30.** Town map (tap-to-travel), investing screen, epilogue and "what if" screens against the real engine; mini-games wired into play; Telugu screenshots; a peer play test; NGO dashboard linked from the end report; demo save for the video.
- **08:30 feature freeze.** Cross play tests, bug fixes, demo video, final report, README with asset credits.
- **Cut list, in order:** second mini-game; Telugu beyond onboarding; Stage 3 depth; town map (chapter map goes straight to the battle); live email (download only); what-if screen (text comparison).